Part One: Choose the correct answer (10 marks)

1	2	3	4	5

1- One of the following matrices is in R.R.E form

A) $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

B) $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

C) $\begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

2- The corresponding linear system for the augmented matrix $\begin{bmatrix} 2 & 1 & 3 & -5 \\ 1 & 2 & 4 & 0 \\ -1 & 6 & 2 & 1 \end{bmatrix}$ is:

A) $\begin{aligned} 2X_1 + X_2 + 3X_3 + 5 &= 0 \\ X_1 + 2X_2 + 4X_3 &= 0 \\ -X_1 + 6X_2 + 2X_3 - 1 &= 0 \end{aligned}$

B) $\begin{aligned} 2X_1 + X_2 + 3X_3 &= 5 \\ X_1 + 2X_2 + 4X_3 &= 0 \\ -X_1 + 6X_2 + 2X_3 &= 1 \end{aligned}$

C) $\begin{aligned} 2X_1 + X_2 + 3X_3 + 5 &= 0 \\ X_2 + 2X_1 + 4X_3 &= 0 \\ -X_1 + 6X_2 + 2X_3 - 1 &= 0 \end{aligned}$

D) $\begin{aligned} 2X_1 + X_2 + 3X_3 + 5 &= 0 \\ X_1 + 2X_2 + 4X_3 &= 0 \\ X_1 - 6X_2 - 2X_3 - 1 &= 0 \end{aligned}$

3- The solution of the linear system with the corresponding augmented matrix $\begin{bmatrix} 2 & 5 & 8 \\ 4 & 10 & 16 \end{bmatrix}$ is:

A) The system has no solution

B) $X_1 = 4$ and $X_2 = 0$

C) $X_1 = 4 - \frac{5t}{2}$ and $X_2 = t;\ t \in \mathbb{R}$

D) $X_1 = 8 - 5t$ and $X_2 = t;\ t \in \mathbb{R}$

4- One of the following statements is false for any symmetric matrix A

A) $tr(2A) = tr(A) + tr(A^T)$.

B) AA^T is symmetric.

C) A is a square matrix.

D) The equation $AX = B$ has only one solution.

5- Let A be a matrix with $det(A) = 7$, and B is the matrix result by multiplying the third row in A by the scalar $\alpha = 3$ and the second column in A by the scalar $\beta = -2$ then $det(B) =$

A) $\frac{-7}{6}$

B) -42

C) 21

D) -14

Part two: Fill the blank. (10 marks)

1- Let $A = \begin{bmatrix} 2 & -7 \\ -3 & 0 \end{bmatrix}$ then $A^{-1} =$ ..... $\begin{bmatrix} 0 & -\frac{1}{3} \\ -\frac{1}{7} & -\frac{2}{21} \end{bmatrix}$ .....

2- Let $B = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 3 & 7 \\ 0 & 0 & b \end{bmatrix} \begin{bmatrix} 2 & 0 & 6 \\ 0 & 5 & 2 \\ 0 & 0 & \frac{1}{25} \end{bmatrix}$ with $det(B) = 10$ then $b =$ ..... 15 .....

3- If $A^{-1} = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix}$ then the solution of the equation $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ is ..... $\begin{bmatrix} 20 \\ 7 \\ 1 \end{bmatrix}$ .....

4- If A is a square matrix satisfies $A^2 + 5A + 6I = I$, then $(A+3I)^{-1} = \ldots 5 \ldots$

5- Let A and B be two symmetric matrix with the same size, then AB is symmetric matrix iff $AB = \ldots BA \ldots$

Part Three: Essay Questions

1- Use Cramer's Rule to solve the linear system (4 marks)

$$x + 2y - z = 5$$
$$2x + y + z = 0$$
$$-x + y + 2y = 1$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow \Delta_1$$

$$\begin{bmatrix} 1 & 5 & -1 \\ 2 & 0 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$\Delta = 1*(2 - (1*1)) - 2(4+1) - 1(2+1)$$
$$= 1 - 10 - 3 = \boxed{-12}$$

$$\Delta_1 = 5(2-1) - 2(0-1) - 1(0-1) = 5 + 2 + 1 = 8$$

$$\Delta_2 = 1(0-2) - 5(4+1) - 1(4-0)$$
$$= -2 - 25 - 4 = -31$$

$$\Delta_3 = 1 + 10 + 0 - (-5 + 4 + 0) = 11 + 1 = 12$$

$$x = \frac{-8}{12}, \; y = \frac{31}{12}, \; z = -1$$

2- Use Gauss-Jordan Method to solve the system (2 marks)

$$x + 3y = 5$$
$$2x + 4y = 8$$

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 4 & 8 \end{array}\right] \Rightarrow \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -2 & -2 \end{array}\right] -2R_1 + R_2$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 1 & 1 \end{array}\right] \frac{R_2}{-2} \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right] -3R_2 + R_1$$

$$(x, y) = (2, 1)$$

3- Find the inverse matrix for $A = \begin{bmatrix} 1 & 3 & 2 \\ -2 & 1 & 4 \\ 0 & 5 & 0 \end{bmatrix}$ (4 marks)

$$A^{-1} = \frac{1}{\det} adj(A)$$

$$= \frac{-1}{40} \begin{bmatrix} -20 & 20 & 0 \\ 0 & 0 & 40 \\ -20 & 20 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} = A^{-1}$$

$$MA = \begin{bmatrix} -20 & 0 & -20 \\ -20 & 0 & -20 \\ 0 & -40 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} -20 & 0 & -20 \\ 20 & 0 & 20 \\ 0 & 40 & 0 \end{bmatrix}$$

$$adj(A) = \begin{bmatrix} -20 & 20 & 0 \\ 0 & 0 & 40 \\ -20 & 20 & 0 \end{bmatrix}$$

$$\det = 1(-20) + 3(0) + 2(-40)$$

$$= -20 - 20 = -40$$